

Find your cybersecurity career path

11 specializations to consider
as you advance your career

INFOSEC[™]



Navigating your cybersecurity career

Are you feeling overwhelmed by the vast landscape of cybersecurity career options? You're not alone! With ever-evolving technology and a dynamic job market, it can be tough to pinpoint the perfect path for your unique skill set and aspirations.

Even if you've been a cybersecurity professional for years, you might be surprised by the number of specializations available. Additionally, some professionals enter cybersecurity from other fields and may not have had exposure to the diverse range of roles within cybersecurity teams.

Let this ebook be your compass, guiding you through the possibilities and helping you find your way toward the perfect cybersecurity career. In the following pages, we'll provide a comprehensive breakdown of 11 in-demand specializations ideal for mid-level cybersecurity professionals. Learn what each job role entails, the related job titles you might find for each role, and the requirements to land those coveted positions.

How the roles were chosen

The cybersecurity job roles in this ebook were assembled based on feedback from our internal Infosec subject matter experts, as well as 20-plus years of feedback working with organizations to help upskill and certify their workforce. We worked with those organizations to find the most common "buckets" of cybersecurity jobs aligned with their hiring and training needs. Those 11 roles are outlined below.

What's in this guide?

- 1 [11 in-demand cybersecurity roles](#)
- 2 [Advice from peers working in those roles](#)
- 3 [Tips to advance your career](#)
- 4 [Resources](#)

Popular cybersecurity careers to consider

It can be hard to evaluate yourself but try to think through questions, like the ones on the right, and find out what it is about working in cybersecurity that you most enjoy. Also, preferences may evolve as you gain experience, so it's important to continually reassess what you enjoy as your career progresses.

Choosing a cybersecurity career requires self-reflection. Consider these questions:

- » Do you thrive in high-pressure situations or prefer a more structured, predictable environment?
- » Would you rather spend time in meetings focused on strategy, managing and reporting to stakeholders, or do you prefer to dive deep into day-to-day tactics and tools?
- » Are you more interested in protecting systems and responding to incidents, or do you want to go on the offensive and learn how malicious actors think?



SOC Analyst

Monitor security systems for suspicious activity and defend against cyberattacks.

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Digital Forensics Analyst

Recover and analyze electronic evidence from cybercrime scenes.

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Penetration Tester

Identify security vulnerabilities and systems by simulating real-world cyberattacks.

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OT Security Practitioner

Safeguard our critical infrastructure, such as power grids and manufacturing plants.

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Security Engineer

Design, implement and maintain the security tools and technologies that keep our data safe.

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Cloud Security Engineer

Ensure the secure configuration and management of sensitive data stored in the cloud.

[Learn More](#)



Security Architect

Design and implement an organization's comprehensive security strategy.

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Information Risk Analyst

Identify, analyze and mitigate potential security threats to an organization's data and systems.

[Learn More](#)



Security Manager

Oversee the overall security posture of an organization.

[Learn More](#)



Privacy Manager

Ensure an organization complies with data privacy regulations and best practices.

[Learn More](#)



Secure Coder

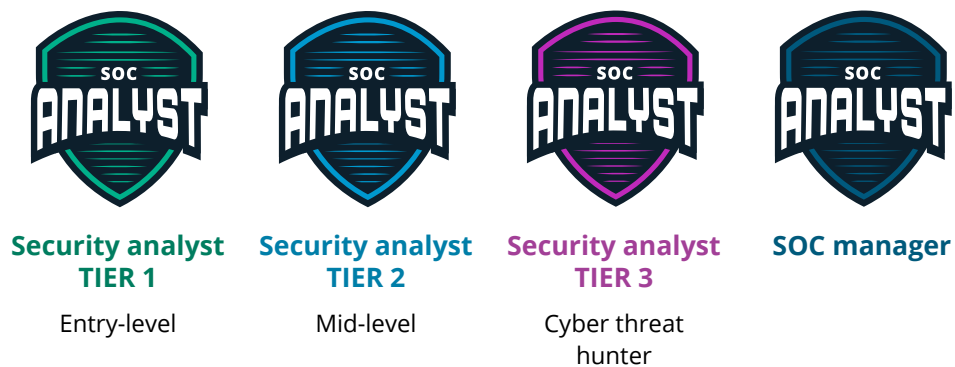
Develop secure software applications that are resistant to vulnerabilities.

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SOC analyst

Often seen as an entry-level position, the SOC analyst role offers a springboard for a thriving cybersecurity career. As you move into mid-level and advanced-level SOC positions, you'll get opportunities to dig deeper into security operations, hone your analytical skills and take on greater responsibility. The specific tasks you perform can vary significantly depending on the organization. A healthcare provider's associate analyst might focus on ensuring patient data is protected, while a financial institution's analyst might specialize in detecting fraudulent transactions.

Below is an example of how your career could progress in such a role:

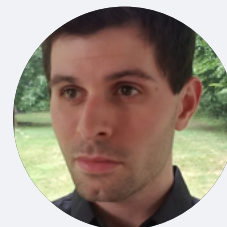


How do I get this role?

- » Familiarity with security incident and event management (SIEM) Tools
- » The ability to interpret and extract meaning from complex security logs
- » Understanding incident response procedures and best practices
- » Consider pursuing these certifications: [CompTIA Security+](#), [CompTIA CySA+](#), [Certified Cyber Threat Hunting Professional \(CCTHP\)](#) and [CertNexus CyberSec First Responder](#)

Mark Viglione on working in a SOC

"There is the technical side, which you have to have. Like analytics, you have to be able to interpret that — and want to do that because you'll be doing a lot of it," says Infosec Skills author Mark Viglione. "And then on the soft skill side, you need to present your findings to your team member, maybe your SOC manager, maybe an IT director who has no idea what some of the technical jargon is and they still need to know."



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Digital forensics analyst

A digital forensics analyst — also known as a computer or cyber forensics analyst or a digital forensics incident response professional — is responsible for extracting and analyzing data from digital devices for investigative purposes. They follow strict chain-of-custody rules and present the retrieved evidence in legally acceptable formats. In smaller organizations, they often wear multiple hats and may go by a more general title, such as computer forensic expert. Larger firms have entire teams focusing exclusively on digital forensics, so you'll likely find the digital forensic analyst title in these organizations.

Below is an example of how your career could progress in such a role:



Digital forensics analyst



Digital forensic investigator



Digital forensics manager



Director of digital forensics

How do I get this role?

- » Strong understanding of computers and operating systems
- » Excellent analytical and problem-solving skills, as well as attention to detail
- » Ability to write and communicate complex technical findings to non-technical audiences
- » Consider pursuing these certifications: [Certified Computer Forensics Examiner \(CCFE\)](#), [Certified Mobile Forensics Examiner \(CMFE\)](#) and [Certified Ethical Hacker \(CEH\)](#).

Amber Schroeder on working as a digital forensics analyst



"Go out and start researching," says Amber Schroeder, Paraben founder and Infosec Skills author. That's the best way to get started and find out if this role is one you want to continue pursuing. Digital forensics professionals often don't know the problem. They have to go out and research each problem, so do the same for your career. Once you decide if it's a career for you, Schroeder says there are opportunities for digital forensics analysts everywhere: midsize and large companies, consulting firms and the public sector.

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Penetration tester

Penetration testers are responsible for planning and performing authorized, simulated attacks within an organization's information systems, networks, applications and infrastructure to identify vulnerabilities and weaknesses. They document their findings in penetration testing reports to advise clients on how to lower or mitigate risk. As their skills grow, they can specialize in different types of pentesting, such as mobile and web application pentesting, or cloud pentesting. The demand for penetration testers is increasing as businesses rely more on technology and face a growing threat landscape.

Below is an example of how your career could progress in such a role:



Penetration
tester



Senior
penetration
tester



Penetration
test lead



Security
architect



Chief
information
security
officer (CISO)

How do I get this role?

- » Strong understanding of computer systems, networks and operating systems
- » Knowledge of hacking techniques and methodologies
- » Experience with penetration testing tools and frameworks.
- » Consider pursuing these certifications: [Certified Ethical Hacker \(CEH\)](#), [CompTIA PenTest+](#), [Certified Cloud Penetration Tester \(CCPT\)](#) and [Certified Mobile and Web Application Penetration Tester \(CMWAPT\)](#).

Keatron Evans on working as a penetration tester



"Problem-solving skills are at the top of the list for penetration testers," Keatron Evans, VP Portfolio and Product Strategy at Infosec. "We can teach you the tools and techniques, but you need to be able to think critically and solve problems."

With some roles, "you're primarily using tools and techniques that are out there in the general public," Evans says. But with others, "You might have to create things from scratch."

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OT security practitioner

Power plants, nuclear reactors, electrical grids: our critical infrastructure is vulnerable to cyberattacks, and it's the job of an OT security practitioner (or ICS security practitioner) to ensure that doesn't happen. An OT security practitioner or ICS/SCADA security engineer monitors, protects and provisions access to ICS systems and devices. Innovations like IoT and cloud technology have brought new avenues of attack — and increased job opportunities for ICS, OT and SCADA security professionals.

Below is an example of how your career could progress in such a role:



OT security analyst



SCADA security engineer



OT security manager

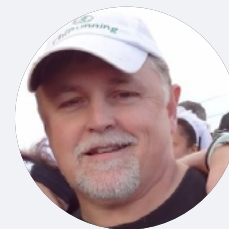


Director of OT security

How do I get this role?

- » Understanding of ICS and automation protocols like SCADA and Modbus
- » Knowledge of cybersecurity principles and network security
- » Familiarity with network security tools and threat detection techniques
- » Consider pursuing these certifications: [ICS Certified Security Professional \(ICSP\)](#), [CompTIA Network+](#) and [CompTIA Security+](#)

Steve Allen on working as an ICS security practitioner



"If you look at cyber war nowadays, one of the big things that these nation states want to do to each other is to try to take down things like the power grid," says Infosec Skills author Steve Allen. Since these bad actors are always hunting, "What you're going to do on a day-to-day basis is make sure that is secure — make sure that there are basically no bad guys getting in."

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Security engineer

A security engineer designs, implements and maintains the security measures of an organization's computer systems and networks. They work closely with other IT professionals to protect against unauthorized access, theft and damage to sensitive information and systems. As a security engineer, you will be responsible for identifying and mitigating potential security threats, as well as developing and implementing security protocols to prevent future breaches. This is a broad role that can vary quite a bit from organization to organization. With experience and additional education, you may move into leadership roles or specialize in a specific area of security engineering.

Below is an example of how your career could progress in such a role:



How do I get this role?

- » Understanding of security threats, vulnerabilities, mitigation strategies and other cybersecurity concepts
- » Experience with security tools and technologies like firewalls, intrusion detection systems and vulnerability scanners
- » Communication and collaboration skills to work effectively with other IT professionals and stakeholders
- » Consider pursuing these certifications: [CompTIA Security+](#), [CompTIA CASP+](#), [ISC2 Certified Information Systems Security Professional \(CISSP\)](#) and [AWS Certified Security – Specialty](#)

Mike Meyers on working as a security engineer



"The security engineer is going to be the person going, 'What are the security issues with our upgrade to this new operating system?'" says Infosec Skills author Mike Meyers. They referee and dissect the security implications of new tools and processes — but he also stresses the importance of soft skills. "Security engineers deal with a lot of people. So good communication skills are so important."

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Information risk analyst

An information risk analyst plays an important role in safeguarding an organization's data and IT systems by conducting objective, fact-based risk assessments on existing and new systems and technologies. They analyze potential threats and vulnerabilities, communicate findings to stakeholders and identify opportunities to improve the organization's risk posture while continually monitoring risk tolerance. They may work as a staff member or a consultant, potentially working for various companies at any given time.

Below is an example of how your career could progress in such a role:



**Junior
information
risk analyst**



**Senior
information
risk analyst**



**Information
risk manager**



**Director of
information risk
management**

How do I get this role?

- » Strong understanding of information security principles, including risk assessment methodologies, data security regulations and common cyber threats
- » Excellent analytical and problem-solving skills
- » Communication and collaboration skills with other IT teams and stakeholders
- » Consider pursuing these certifications: [ISACA Certified Information Systems Auditor \(CISA\)](#), [ISC2 Certified in Governance Risk and Compliance \(CGRC\)](#), [ISACA Certified in Risk and Information Systems Control \(CRISC\)](#) and [Certified CMMC Professional \(CCP\)](#)

Chris Stevens on working as an information risk analyst



"You're there to help the organization understand risk on all levels — to identify, analyze and help owners make informed decisions on whether to accept, avoid, transfer or mitigate that risk," says Infosec Skills author Chris Stevens. "Risk touches everything an organization does. You need risk managers to look at activities, measure inherent risks, and reduce those risks without impacting profitability and survivability."

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Cloud security engineer

A cloud engineer is responsible for securely deploying infrastructure into the company's cloud provider. They will also set up guardrails to ensure others do not deploy insecure infrastructure. With more companies migrating to cloud platforms, cloud skills are important even if you aren't pursuing a cloud security engineer role. According to the ISACA State of Cybersecurity Report, cloud computing is the biggest skills gap among cybersecurity professionals.

Below is an example of how your career could progress in such a role:



Cloud
operations
administrator



Cloud security
engineer



Cloud solutions
architect



Director
of cloud
infrastructure

How do I get this role?

- » Understanding of cloud security concepts like access control, encryption and identity and access management (IAM)
- » Familiarity with cloud providers like AWS, Microsoft Azure and Google Cloud Platform
- » Knowledge of compliance regulations and how they affect secure cloud architectures
- » Consider pursuing these certifications: [ISC2 Certified Cloud Security Professional \(CCSP\)](#), [AWS Solutions Architect](#), [AWS Certified Security](#) and [Azure Security Engineer](#)

Joe South on working as a cloud security engineer



"It definitely cannot be your first job," says Infosec Skills Author Joe South. "Cloud security spans across just about every single domain of security. A cloud security engineer needs to have experience in several different domains. I'm not just talking about two domains. I'm talking about three, four, five domains across security. The first step is, obviously, the experience." Josh recommends getting certified with the cloud provider your current company uses to gain foundational knowledge.

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Security manager

Security managers are the strategists and leaders who oversee an organization's entire security posture. They're responsible for developing, implementing and maintaining a comprehensive security program that protects the organization's data, applications, systems and facilities from cyberattacks, physical threats and other security risks. Security managers work closely with IT teams, physical security teams and other departments to ensure a coordinated security approach.

Below is an example of how your career could progress in such a role:



Security manager



Director of
information security



Chief information
security officer (CISO)

How do I get this role?

- » Understand up-to-date security best practices, frameworks and methodologies, such as the NIST Cybersecurity Framework (CSF) and ISO 27001
- » Excellent communication and interpersonal skills to work effectively with different teams within an organization and collaborate with external vendors
- » Strong leadership and mentoring skills
- » Consider pursuing these certifications: [ISC2 Certified Information Systems Security Professional \(CISSP\)](#), [ISACA Certified Information Security Manager \(CISM\)](#), [IAPP Certified Information Privacy Manager \(CIPM\)](#) and [PMI Project Management Professional \(PMP\)](#)

Cicero Chimbanda on working as a security manager



"There are two ways that people take the security manager track," says Infosec Skills author Cicero Chimbanda. "You have individuals that are technical in nature, or you can have individuals that become a manager because they have really good soft skills and people skills. They have good business acumen, good project management acumen. They're not necessarily technical, but they know how to connect the dots. They manage subject matter experts."

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Security architect

A security architect is responsible for designing an organization's security infrastructure and implementing policies and guidance to help minimize potential cyber risk. This position is generally held by someone with a solid security background and years of experience working in a position like security engineer. They can be some of the most versatile people in the organization. Often, security architects are part developer, part engineer and part security expert and have extensive experience in various domains of IT.

Below is an example of how your career could progress in such a role:



Information
security
architect



Principal
security
architect



Cybersecurity
consultant



Chief security
officer (CSO)

How do I get this role?

- » Experience with networking, cryptography and operating systems, as well as an understanding of network, cloud and application security
- » Understanding of best practices, frameworks and methodologies, such as the NIST Cybersecurity Framework (CSF) and CIS Controls
- » Ability to assess complex systems, identify potential risks and develop solutions to mitigate security threats
- » Consider pursuing these certifications: [ISC2 Certified Information Systems Security Professional \(CISSP\)](#), [ISC2 Certified Cloud Security Professional \(CCSP\)](#) and [CompTIA Advanced Security Practitioner \(CASP+\)](#)

Leighton Johnson on working as a security architect



"Start working toward security architect, whether you're a security analyst, a pentester or something else," says Infosec Skills author Leighton Johnson. "But you're going to have to do it from two perspectives. You're going to have to be doing it from the security perspective. And you'll also have to be exposed to where they exist in the business and what the business is doing."

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Privacy manager

A privacy manager, or privacy officer, helps operate and improve an organization's privacy practices and programs. Responsibilities will vary because this is an evolving and expanding field that intersects with security, law, technology and compliance. Typical duties include keeping current on laws and regulations, updating corporate security policies, conducting privacy impact assessments (PIAs) and managing consumer inquiries and data requests.

Below is an example of how your career could progress in such a role:



Privacy analyst



Privacy manager



Data protection officer (DPO)



Chief privacy officer (CPO)

How do I get this role?

- » Understanding of data privacy frameworks, standards and regulations, such as GDPR, HIPAA and CPRA
- » Ability to identify, assess, and manage privacy risks through assessments and risk mitigation strategies
- » Strong communication skills to collaborate and educate various stakeholders
- » Consider pursuing these certifications: [IAPP Certified Information Privacy Manager \(CIPM\)](#), [IAPP CIPP/US](#), [IAPP CIPP/Europe](#) and [ISACA Certified Data and Privacy Solutions Engineer \(CDPSE\)](#)

Chris Stevens on working as a privacy manager



"We're always talking about certs," says Infosec Skills author Chris Stevens.

"That gives credibility to you as a privacy professional. Many of us start out with one of the policy certs, whether that's a Certified Information Privacy Professional for the U.S. or Europe. Learn the nuances of the law to write policies, procedures and standards. Understand how to become an effective communicator. Privacy managers have to be good listeners; they have to be forward-thinking."

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Secure coder

Secure coders are the frontline defenders against software vulnerabilities. They know how to write code like other software developers but do so with security in mind. They employ techniques to prevent common attacks like SQL injection and cross-site scripting (XSS) in all parts of the software development lifecycle (SDLC), from writing the code to deploying it to production. They work with other developers and DevOps engineers to ensure applications are built with robust security features.

Below is an example of how your career could progress in such a role:



Software
developer



Secure code
analyst



Application
security
engineer



Application
security
architect

How do I get this role?

- » Proficient in various programming languages
- » Understanding of secure coding practices and common vulnerabilities, such as the OWASP Top 10
- » Strong problem-solving and analytical skills, as well as attention to detail
- » Consider pursuing these certifications: [ISC2 Certified Secure Software Lifecycle Professional \(CSSLP\)](#), [CertNexus Cyber Secure Coder](#) and [Certified Ethical Hacker \(CEH\)](#).

John Wagnon on working as a secure coder



"You need to know how to program," says Infosec Skills author John Wagnon, "Those programming languages — whether it's Go or Java or Python or whatever — I would pick one if you're just getting started." Wagnon also recommends contributing to open-source projects to gain experience. "Pick one of them, grab it and start working on it. Then you can take that to an employer to showcase your skills."

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Advancing your cybersecurity career

This ebook has explored 11 of the most common specializations within cybersecurity:

- » SOC analyst
- » Digital forensics analyst
- » Penetration tester
- » OT security practitioner
- » Security engineer
- » Information risk analyst
- » Cloud security engineer
- » Security manager
- » Security architect
- » Privacy manager
- » Secure coder

Each of these specializations offers a unique career path with opportunities for advancement. As you gain experience and expertise, you can progress to leadership roles, specialize in a particular niche or even move laterally to a new role using the transferable skills you've gained.

There can be quite a bit of variety within each role. Penetration testing firms, for example, encompass a range of specialties: Depending on your interests and skills, you might focus on web application testing, social engineering or cloud security — or remain a generalist. This allows you to create your own unique path within a broader specialization as you discover your passions and strengths.

We encourage you to explore the different cybersecurity career paths and discover what you're passionate about. Don't be afraid to experiment and try new things. Many of the skills you develop in one area of cybersecurity are transferable to others. The important thing is to keep learning.

Thomas Pace on finding your career path

Remember that even as you pursue a specialty, you're not locked into a career path, so don't be paralyzed by choosing. Thomas Pace, founder of NetRise, offers this simple advice: "Go do what you want to do." Just get started, and if you find out you don't like something, that's a good thing. The most important part is to see what you like and continue building your skills.



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Tips for moving up the career ladder

Successfully moving up the cybersecurity career ladder requires continuous learning and development. Here are some strategies to employ to give yourself a leg up:



Continuously expand your skillset

The [ISC2 Cybersecurity Workforce Study](#) found that organizations with paid training reimbursement programs were “far better prepared to weather times of economic uncertainty.” If your company offers training programs, take advantage of them. It’s a win-win for both you and your organization.



Stay abreast of industry trends

While exploring specialization paths, consider aligning your interest with the industry’s evolving needs. The same study found that cloud computing security, risk management, security analysis, security engineering and application security are the top skills hiring managers look for.



Build your network

A different [study by ISSA](#) found that 62% of professionals ranked networking with other cybersecurity professionals as the most helpful action for career progression. Attend industry events, connect with professionals online and participate in relevant communities to build strong relationships and gain valuable insights.



Earn industry certifications

Industry-recognized certifications from organizations like CompTIA, ISC2, ISACA, AWS, Cisco and others validate your knowledge and skills, significantly enhancing your job prospects.

Alicia Olson on the importance of being uncomfortable



Alicia Olson, VP of Communications at Optiv, offers her personal mantra as the best career advice she received: “Get comfortable with being uncomfortable.” This means challenging yourself by taking on stretch projects or goals that are outside your comfort zone. By constantly seeking new challenges and opportunities for growth, you can advance your career.

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Additional resources

- » [Infosec Resource Center](#)
- » [Infosec webcasts and events](#)
- » [Cyber Work Podcast](#)
- » [Infosec YouTube channel](#)

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About Infosec

Infosec's mission is to put people at the center of cybersecurity. Through role-guided security training, our platforms — Infosec IQ, Infosec Skills and Infosec Boot Camps — help individuals and organizations protect their data, mitigate risk and empower employees. From security awareness for your accounting team to secure coding training for your developers, we'll help you deliver the right security education to protect your employees and organization.

Learn more at infosecinstitute.com.